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Super-Agers' Brains Have a Special Ability, New Study Suggests

The findings may help explain why this group has such exceptional memory.

结果状语，现在分词

Many people's brains deteriorate as they age, becoming riddled with malfunctioning proteins that result in cell death and the loss of memory and cognition. But other people's brains remain almost perfectly intact, their thinking as sharp at 80 as it was in their 50s.

A paper published Wednesday in the journal Nature provides a new potential explanation for this discrepancy, and it taps into one of the hottest debates in neuroscience: whether human brains can grow new neurons in adulthood, a phenomenon called neurogenesis. 神经发生

The study found that so-called super-agers — people 80 and up who have the memory ability of someone 30 years younger — had roughly twice as many new neurons as older adults with normal memory for their age, and 2.5 times more than people with Alzheimer's disease. The research focused on an area of the brain called the hippocampus, which is important for learning and memory and is thought to be the primary birthplace of new neurons.

"This paper shows biological proof that the aging brain is plastic," even into a person's 80s, said Tamar Gefen, an associate professor of psychiatry and behavioral sciences at the Northwestern University Feinberg School of Medicine, who contributed to the research.

To look for neurogenesis in older adults, the scientists first tried to detect signs of it in the autopsied brains of young adults, age 20 to 40, who died with normal cognition. They identified genetic markers for three key types of cells: neural stem cells, neuroblasts and immature neurons.

"It's almost like neural stem cells are babies, neuroblasts are kind of teenagers and immature neurons are kind of almost adults," said Orly Lazarov, a professor of neuroscience at the University of Illinois at Chicago College of Medicine, who led the research. The presence of all three types could suggest that stem cells are active and dividing in the brain and that those new baby cells are maturing into adult neurons.

Next, the scientists searched for these same three cell types in the brains of four groups of older adults: people with normal cognition, those with mild cognitive impairment, those with Alzheimer's and super-agers, all of whom had donated their brains to be studied after they died. Each group had signs of all three cell types, but the amounts differed dramatically among them and appeared to relate to people's cognition at the time of death.

The super-agers had substantially more immature neurons in their hippocampi — not only compared with the other older adults, but with the young adults, as well. The super-agers' immature neurons also had unique genetic and epigenetic characteristics that the researchers think made them resilient to aging.

"Super-aging happens not only because there's more of these young cells, but because there is a type of genetic programming" that allows for their preservation, Dr. Gefen said.

Dr. Bryan Strange, a professor of clinical neuroscience at the Polytechnic University of Madrid who studies a different group of super-agers, said neurogenesis could help to account for other unique aspects of super-ager brains, including that the hippocampus is often much larger than it is in typical older adults.

But he pointed out that super-agers have other brain differences, like more volume in areas that don't experience neurogenesis and greater connectivity between brain regions, that can't be explained by the new findings.

The research uncovered something interesting about people in the Alzheimer's group, too. They actually had more neural stem cells compared with the other older adults, but many fewer neuroblasts and immature neurons.

"If you have normal neurogenesis, you gradually lose the stem cells," said Hongjun Song, a professor of neurological sciences at the University of Pennsylvania Perelman School of Medicine who researches neurogenesis but was not involved in the study. One interpretation of the new finding is that, in Alzheimer's, neurogenesis is disrupted and the stem cells get turned off and aren't able to progress to the next stage of development, so the stem cell pool is preserved.

"If that's true, that's really opened up a new direction for the field" to potentially treat Alzheimer's by reactivating the dormant stem cells, Dr. Song said.

Not everyone is convinced of the new findings. Shawn Sorrells, an associate professor of neuroscience at the University of Pittsburgh who has also researched neurogenesis, said that the scientists' goal of mapping "how the hippocampus changes with aging and changes differently in people who age differently is fantastically interesting and important."

But Dr. Sorrells worries the study suffers from some of the same methodological flaws and assumptions as other research on neurogenesis. He added that he would like to see the findings validated using other techniques.

Experts agree that babies and young children are able to generate new neurons in the brain, as are several species of adult animals. But many think it's still not clear if human adults have the same ability.

There are numerous studies providing evidence on both sides, and the results are often influenced by what methods researchers used.

This latest study most likely won't settle the debate, but it does give scientists new leads to pursue. For her part, Dr. Lazarov is now trying to understand how super-agers' special immature neurons relate to the group's superior memory, and if it might be possible to capture some of that activity in a drug to help others stay sharper longer.